

Third-Person Perspective Avatar Embodiment in Augmented Reality: Examining the Proteus Effect on Physical Performance

Riku Otono*
NAIST, Japan

Naoya Isoyama†
NAIST, Japan

Hideaki Uchiyama‡
NAIST, Japan

Kiyoshi Kiyokawa§
NAIST, Japan

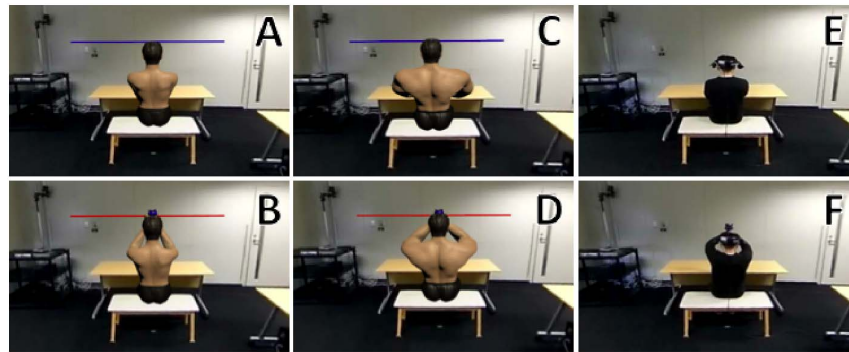


Figure 1: User's view with the non-muscular avatar while lifting down (A) and up (B), user's view with the muscular avatar while lifting down (C) and up (D), the original view while lifting down (E) and up (F).

ABSTRACT

Embodiment in augmented reality (AR) is applicable to various fields such as exercise and education. However, full-body embodiment in AR is still challenging to implement due to technical problems such as low body tracking accuracy. Therefore, the study on the impact of an avatar in AR on user performance is limited. We implemented an AR embodiment system and investigated its impact on user physical performance. The system allows users to see their avatar instead of their real body from a third-person perspective. The results show that a muscular avatar improves user physical performance during and after controlling the avatar.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed/augmented reality; Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI

1 INTRODUCTION

Avatars are widely used in various computer systems such as virtual conferences and video games. A rich body of virtual reality (VR) research has shown that an embodied avatar may change user attitude, behavior, and performance. This phenomenon is called the Proteus effect [6]. For instance, taller avatars make their users behave more confidently in a negotiation task. On the other hand, to the best of our knowledge, no study on the Proteus effect was conducted in augmented reality (AR).

This is probably because of the challenges surrounding AR embodiment representation [1]. Unlike VR, a user's real body is visible in AR and it would distract users from focusing on their avatar.

The user's body becomes conspicuous when tracking errors occur and the avatar's shape differs from that of the user. The main con-

tributions of this work are 1) the implementation of a third-person perspective (3PP) embodiment system in AR and 2) validation of the Proteus effect in 3PP in AR.

First, we aim to provide users full body embodiment in AR with a completely diminished user body and an avatar synchronized with the user's motion. To achieve this goal, we combine a 3PP, a pre-captured 360 image, and tracker-based motion tracking. A 3PP will allow the user to see the avatar at all times, without any equipment showing the avatar appearance like a virtual mirror. A virtual mirror is often used in VR but it will block the user's view and would be impractical in AR. A previous study on a 3PP AR embodiment found that a synchronized avatar, which appeared in front of the user, provided the sense of agency [5]. While the participants could see their own body in [5], our system allows seeing one's self-avatar without one's real body.

Second, we investigate the Proteus effect of avatar body shape from a 3PP in AR on physical performance of male participants. Thus, the impacts of two body shapes (a non-muscular and a muscular avatars) on physical performance are compared during and after controlling these bodies. Lin et al. previously examined the Proteus effect induced by a muscular avatar using the first-person perspective (1PP) in VR [3]. They showed that the Proteus effect was confirmed for female participants who were embodied in a muscular avatar, but not for male participants. Thus, it remains unanswered if the Proteus effect is induced for male users by a muscular avatar, in AR, or from a 3PP.

2 SYSTEM

Our proposed system enables users to observe their avatars from a 3PP in AR. A user wears a head-mounted display (HMD) (HTC VIVE Pro) and observes their avatar from behind it in a real-time 360 video (see Fig. 1). A part of the 360 image is replaced with a static image captured in advance without the user in the image. The size of the static image is set beforehand to cover the user's real body. The 360 images are captured by a Ricoh Theta V positioned at 1.5 meters above the ground and about 2 meters behind the user.

Both a non-muscular avatar and a muscular avatar were created using Daz3D. The avatars move synchronously with user movement measured by a motion tracker (HTC VIVE Tracker) and the HMD.

*e-mail: otono.riku.ot6@is.naist.jp

†e-mail: isoyama@is.naist.jp

‡e-mail: hideaki.uchiyama@is.naist.jp

§e-mail: kiyokawa@is.naist.jp

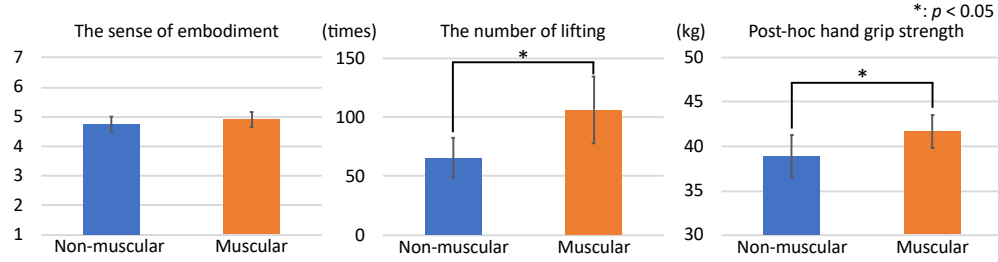


Figure 2: Sense of embodiment measured on a Likert scale from 1 to 7 (left), the number of up-down lifts of the dumbbell (middle), and post-hoc hand grip strength (kg) (right).

3 EXPERIMENT

3.1 Overview

The purpose is to examine if the Proteus effect occurs in AR using the proposed system. The study followed a within-subjects design to compare a non-muscular and a muscular avatar body shape. The conditions were counterbalanced with a Latin-square design. 16 male participants aged between 22 and 34 ($M=23$, $SD=2.87$) participated in this study. All but one of the participants were right-handed. All participants had experienced VR at least once.

We assumed, compared to the muscular avatar, that the non-muscular avatar body appearance would be closer to the expected participants' one. Thus, we expected that **H1) compared to the muscular avatar, participants would feel a higher sense of embodiment (SoE) for the non-muscular avatar.** The SoE refers to how much users feel that they own, control, and are inside in their avatar [2].

As suggested by similar VR research [3], we expected that **H2) compared to the non-muscular avatar, the muscular avatar would improve user's physical performance while controlling the avatar.**

Previous studies have also shown that the Proteus effect may be maintained even after the avatar embodiment has ended [3]. Thus, we assumed that **H3) compared to the non-muscular avatar, the muscular avatar would improve user's physical performance after controlling the avatar.**

3.2 Procedure

After the participants filled in a pre-questionnaire on their body weight or VR experience, they sat on the center of a bench and put on the HMD. During the AR experiment, the participants could see their avatar in a real-time 360 video of the real environment where their real body was removed. They were instructed to lift up and down a 5 kg dumbbell (Umi Dumbbell) five times as a practice. Then, the participants were asked to lift it up and down every five seconds as many times as possible.

After the participants decided to stop lifting the dumbbell, they took off the HMD and took a break for two minutes. Then, the participants left the bench and their hand-grip strength was measured with a dynamometer (N-FORCE HG-251). Afterwards, the participants answered a questionnaire evaluating their SoE with the questionnaire [4], which consists of 16 questions (7-point Likert scale) with four sub-scales (appearance, response, ownership, and multi-sensory). After that, the participants repeated the task using the other avatar and filled in the questionnaire about the SoE again.

3.3 Results

The SoE was analyzed using a Wilcoxon signed-rank test to verify **H1**. There was no significant differences between the two body shape conditions in terms of SoE ($W=52$, $p=0.43$) (see Fig. 2 (left)).

The number of up-down lifts of the dumbbell was analyzed using a Wilcoxon signed-rank test to verify **H2**. As we expected, the

number of up and down lifts of the dumbbell was significantly larger in the muscular avatar condition than in the non-muscular avatar condition ($W=25$, $p=0.02$) (see Fig. 2 (middle)). Hence the Proteus effect was induced while being embodied in the muscular avatar.

The post-hoc hand-grip strength was confirmed to follow a normal distribution, thus the data was analyzed using a paired t-test to verify **H3**. As we expected, post-hoc hand grip strength was significantly higher after being embodied in the muscular avatar than in the non-muscular avatar ($t(15) = -2.84$, $p=0.01$) (see Fig. 2 (right)). Therefore, the Proteus effect was maintained even after the end of the embodiment phase in terms of hand-grip strength.

4 DISCUSSION

H2 and **H3** were supported, thus it was showed that the muscular avatar from a 3PP in AR improves user physical performance during and after controlling the avatar. However, there are some limitations in this work. First, the participants were instructed to lift up and down the dumbbell as many times as possible in each task. Thus, the duration of embodiment is different for each participant. Furthermore, no female participated in this experiment, thus the Proteus effect from a 3PP in AR for female users remains unanswered.

5 CONCLUSION

In this article, a 3PP AR system was proposed using a synchronized avatar integrated into a real-time 360 video where their real body is completely diminished. We also validated the Proteus effect in a 3PP in AR using the proposed system. The results show that a muscular avatar from a 3PP in AR improves male user physical performance both during and after embodiment. In the future, we will conduct a follow-up study with female users and 1PP.

ACKNOWLEDGMENTS

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REFERENCES

- [1] A. Genay, A. Lécuyer, and M. Hachet. Being an avatar "for real": a survey on virtual embodiment in augmented reality. *TVCG*, PP:1–1, 07 2021.
- [2] K. Kiltner, R. Groten, and M. Slater. The sense of embodiment in virtual reality. *Presence*, 21(4):373–387, 2012.
- [3] J.-H. T. Lin, D.-Y. Wu, and J.-W. Yang. Exercising with a six pack in virtual reality: Examining the proteus effect of avatar body shape and sex on self-efficacy for core-muscle exercise, self-concept of body shape, and actual physical activity. *Frontiers in Psychology*, 12:4580, 2021.
- [4] T. C. Peck and M. Gonzalez-Franco. Avatar embodiment: a standardized questionnaire. *Frontiers in Virtual Reality*, 1:44, 2021.
- [5] N. Rosa, J.-P. van Bommel, W. Hürst, T. Nijboer, R. C. Velkamp, and P. Werkhoven. Embodying an extra virtual body in augmented reality. In *IEEE VR*, pp. 1138–1139, 2019.
- [6] N. Yee and J. N. Bailenson. The proteus effect: The effect of transformed self-representation on behavior. *Human Communication Research*, 33:271–290, 2007.